Tighter Convergence Constants for Multi-Agent Policy Gradient via Precision Weighting and Opponent-Shaping Basin Enlargement

Anonymous Author(s)

Abstract

We establish two orthogonal improvements to policy gradient (PG) convergence in general stochastic games, building on the framework of Giannou et al. Giannou et al. [2022]. **(1) Precision-weighted PG (PW-PG):** when agents face heterogeneous data quality, scaling each agent's gradient update by its inverse-variance weight provably improves the convergence constant by a factor of $\mathrm{HM}(\sigma^2)/\mathrm{AM}(\sigma^2) \leq 1$ (Theorem 2), where HM and AM are the harmonic and arithmetic means of the noise variances. The improvement is largest when data quality is most heterogeneous. **(2) LOLA-PG basin enlargement:** under a *spectral reinforcement condition* on the opponent-shaping Hessian—satisfied in zero-sum and negative-cross-Hessian games—LOLA-type opponent shaping strictly enlarges the basin of attraction, increasing the effective SOS parameter from μ to $\mu + \lambda\mu_H$ (Theorem 6). **(3) Combined PW-LOLA-PG:** the two improvements are *orthogonal* in the Lyapunov analysis, composing to simultaneously tighten the variance prefactor and enlarge the convergence region (Theorem 8). Experiments on Matching Pennies and Battle of the Sexes validate both results: PW-PG achieves up to 67% variance reduction (heterogeneity ratio 10:1), and LOLA-PG extends the convergence basin by 15–40% in zero-sum settings.

1 Introduction

Giannou, Lotidis, Mertikopoulos, and Vlatakis-Gkaragkounis Giannou et al. [2022] proved that policy gradient (PG) methods converge to stable Nash equilibria in general stochastic games, with rate $\mathcal{O}(C/n^q)$ for REINFORCE-type estimators. Their result left open two practical questions:

1. *What happens when agents observe their environments with unequal fidelity?* In realistic MARL deployments—federated learning, heterogeneous robot teams, decentralised market agents— different agents accumulate experience at different rates, leading to heterogeneous gradient noise. Standard PG treats all agents uniformly, wasting information.

2. *Can explicit opponent modelling (LOLA Foerster et al. [2018]) be given convergence guarantees?* LOLA has strong empirical results in iterated games but lacks theoretical grounding in general stochastic games.

We answer both questions, with quantitative guarantees on the improvement. Our contributions connect two bodies of work that have remained separate: the convergence theory of Giannou et al. [2022] and the opponent-shaping literature Foerster et al. [2018], Letcher et al. [2019].

Related work. Global convergence of PG in structured games (potential games, zero-sum) has been studied in Leonardos et al. [2022], Daskalakis et al. [2020]. The n-sided PL condition of Chao and Etesami [2026] gives global convergence under a generalised gradient domination assumption. Convergence of LOLA was studied in Letcher et al. [2019] for normal-form games via game-theoretic analysis, but without the measure-theoretic stochastic stability argument of Giannou et al. [2022]. Our work extends the latter to cover precision-weighted updates and opponent shaping simultaneously. Inverse variance weighting in statistics Cochran [1954] and federated learning Li et al. [2020] motivates our data-quality model; we provide the first convergence bound in the MARL context.

2 Preliminaries

Stochastic game and policy gradient. We adopt the framework of Giannou et al. [2022] exactly. A stochastic game $\mathcal{G} = (\mathcal{S}, \mathcal{N}, \{\mathcal{A}_i\}, P, \zeta, \rho)$ has N agents, state space $\mathcal{S}$, action spaces $\mathcal{A}_i$, Markov transition kernel P, and initial distribution ρ. Agent i's stationary policy $\pi_i \in \Pi_i = \Delta(\mathcal{A}_i)^{\mathcal{S}}$ yields value $V_{i,\rho}(\pi)$. Write $v_i(\pi) = \nabla_i V_{i,\rho}(\pi)$ for agent i's policy gradient and $v(\pi) = (v_i(\pi))_{i\in\mathcal{N}}$.

The standard PG update is $\pi_{n+1} = \mathrm{proj}_{\Pi}(\pi_n + \gamma_n \hat{v}_n)$, where $\hat{v}_n = v(\pi_n) + U_n + b_n$ decomposes into the true gradient, zero-mean noise ($\mathbb{E}[U_n \mid \mathcal{F}_n] = 0$, $\mathbb{E}[\|U_n\|^2 \mid \mathcal{F}_n] \le \sigma_n^2$), and bias ($\|b_n\| \le B_n$).

Key result we build on.

Theorem 1 (Giannou et al. Giannou et al. [2022], Thms. 1–2)**.** *Let π^* be a* second-order stationary (SOS) *Nash policy with parameter $\mu > 0$: $\langle v(\pi), \pi - \pi^* \rangle \le -\mu\|\pi - \pi^*\|^2$ near π^*. Under standard step-size and noise decay conditions, PG converges from a neighbourhood $\mathcal{U}$ of π^* with $\mathbb{P}(\pi_n \to \pi^* \mid \pi_1 \in \mathcal{U}) \ge 1 - \delta$, at rate $\mathbb{E}[\|\pi_n - \pi^*\|^2 \mid \mathcal{E}] = \mathcal{O}(C/n^q)$, with $q = p - 2\ell_\sigma \in (1/2, 1)$.*

The constant C and the neighbourhood $\mathcal{U}$ (which determines the basin probability $p = \nu(\mathcal{U})$ in random-restart analyses) are controlled by two levers: the gradient variance σ_n^2 (enters C) and the SOS parameter μ (enters $\mathcal{U}$ through the Lyapunov stability radius). Our two results improve each lever independently.

3 Precision-Weighted Policy Gradient

Heterogeneous data model. In many MARL settings, agents have access to different amounts of data. We model this via heterogeneous noise levels: agent i's gradient estimator has conditional variance $\mathbb{E}[\|U_{i,n}\|^2 \mid \mathcal{F}_n] \le \sigma_{i,n}^2$, where $\sigma_{i,n}^2 = C/\tau_{i,n}$ and $\tau_{i,n}$ is agent i's effective sample size at episode n. In standard PG, all agents use the same step-size γ_n, so the total gradient variance is $\sigma_n^2 = \sum_i \sigma_{i,n}^2$.

Precision weighting. Scale each agent's update by a precision weight $w_{i,n} \propto 1/\sigma_{i,n}^2 = \tau_{i,n}/C$, normalised so that $\sum_i w_{i,n} = N$. The *precision-weighted PG (PW-PG)* update is

$$\pi_{i,n+1} = \mathrm{proj}_{\Pi_i}\big(\pi_{i,n} + \gamma_n w_{i,n} \hat{v}_{i,n}\big). \tag{1}$$

Theorem 2 (PW-PG variance reduction)**.** *Under the heterogeneous noise model above, PW-PG satisfies all conditions of Theorem 1 with effective total variance*

$$\sigma_{w,n}^2 = \frac{\mathrm{HM}(\sigma_n^2)}{\mathrm{AM}(\sigma_n^2)} \cdot \sigma_n^2 \;=\; \frac{N^2}{\sum_i \sigma_{i,n}^{-2}} \cdot \frac{1}{N \sum_i \sigma_{i,n}^2} \;\le\; \sigma_n^2, \tag{2}$$

where $\mathrm{HM}(\sigma^2)$ and $\mathrm{AM}(\sigma^2)$ are the harmonic and arithmetic means of $(\sigma_{1,n}^2, \ldots, \sigma_{N,n}^2)$. Consequently, the convergence constant improves: $C_w = (\mathrm{HM}/\mathrm{AM}) \cdot C \le C$. Equality holds iff all agents have equal noise variance.

Proof sketch. With weights $w_{i,n} = N\sigma_{i,n}^{-2}/\sum_j \sigma_{j,n}^{-2}$, the weighted noise $\xi_{i,n} = w_{i,n} U_{i,n}$ satisfies $\mathbb{E}[\|\xi_{i,n}\|^2 \mid \mathcal{F}_n] \le w_{i,n}^2 \sigma_{i,n}^2$. Summing over i:

$$\sigma_{w,n}^2 = \sum_i w_{i,n}^2 \sigma_{i,n}^2 = N^2 \left(\sum_j \sigma_{j,n}^{-2}\right)^{-2} \sum_i \sigma_{i,n}^{-2} = \frac{N^2}{\sum_i \sigma_{i,n}^{-2}}.$$

The ratio to $\sigma_n^2 = \sum_i \sigma_{i,n}^2$ equals $\mathrm{HM}(\sigma^2)/\mathrm{AM}(\sigma^2) \le 1$ by the AM-HM inequality (Cauchy–Schwarz). The drift term $\sum_i w_{i,n} v_i(\pi_n)$ preserves the SOS condition since $w_{i,n} > 0$ and $\sum_i w_{i,n} = N$ (the weighting is a positive rescaling of each agent's contribution). All other conditions of Theorem 1 are inherited. □

Remark 3 (Magnitude of improvement)**.** *For $N = 2$ agents with variance ratio $r = \sigma_1^2/\sigma_2^2$, $\mathrm{HM}/\mathrm{AM} = 4r/(1+r)^2$. At $r = 1$: no improvement. At $r = 10$: $\mathrm{HM}/\mathrm{AM} \approx 0.33$, a 67% reduction in the convergence constant. The improvement is monotone in the heterogeneity of the agent population.*

4 LOLA-PG and Basin Enlargement

LOLA update. Foerster et al. Foerster et al. [2018] proposed learning with opponent-learning awareness (LOLA), in which agent i anticipates agent j's next gradient step when computing its own update. In policy-gradient form:

$$\pi_{i,n+1} = \mathrm{proj}_{\Pi_i}\big(\pi_{i,n} + \gamma_n(\hat{v}_{i,n} + \lambda_n \cdot \mathrm{OS}_i(\pi_n))\big), \tag{3}$$

where $\mathrm{OS}_i(\pi) = \sum_{j\neq i}(\partial V_i/\partial \pi_j)\cdot(\partial \pi'_j/\partial \pi_i)$ is the opponent-shaping term, with $\pi'_j = \pi_j - \eta_j \nabla_{\pi_j} V_{j,\rho}(\pi)$.

Lemma 4 (Bias absorption; Giannou et al. [2022] Lemma A.1 extended)**.** *The LOLA gradient is $\hat{v}_n^{\mathrm{LOLA}} = v(\pi_n) + U_n + b_n^{\mathrm{LOLA}}$, where $\|b_n^{\mathrm{LOLA}}\| \le B_n + \lambda_n G$ and $G = \sup_\pi \|\mathrm{OS}(\pi)\| < \infty$. With annealing $\lambda_n = \bar{\lambda}/n^r$, $r > 1-p$, the modified bias satisfies the required decay at rate $\min(\ell_b, r)$.*

The spectral reinforcement condition. Near a Nash policy π^*, the LOLA gradient field linearises as $v^{\mathrm{LOLA}}(\pi) \approx (\mathrm{Jac}_v(\pi^*) + \lambda H(\pi^*))(\pi - \pi^*)$, where $H = \nabla_\pi \mathrm{OS}(\pi)|_{\pi^*}$ is the *opponent-shaping Hessian*. Let $S_H = (H + H^\top)/2$ be its symmetric part.

Definition 5 (Spectral reinforcement)**.** *H is* spectrally reinforcing *at π^* if $S_H \prec 0$, with $\mu_H := -\lambda_{\max}(S_H) > 0$.*

Theorem 6 (Basin enlargement under spectral reinforcement)**.** *Let π^* be SOS with parameter μ under standard PG. If $H(\pi^*)$ is spectrally reinforcing with parameter $\mu_H > 0$, then under LOLA-PG* (3) *with constant $\lambda > 0$, the effective SOS parameter is*

$$\mu_{\mathrm{LOLA}} = \mu + \lambda\mu_H > \mu. \tag{4}$$

The convergence basin $\mathcal{U}_{\mathrm{LOLA}} \supseteq \mathcal{U}_{\mathrm{std}}$, with Lyapunov contraction factor $(1 - 2\mu_{\mathrm{LOLA}}\gamma_n)$ vs. $(1 - 2\mu\gamma_n)$.

Proof. For $\pi \in \mathcal{U}_{\mathrm{std}}$,

$$\begin{aligned}\langle v^{\mathrm{LOLA}}(\pi),\, \pi - \pi^*\rangle &= \langle v(\pi),\, \pi - \pi^*\rangle + \lambda\langle \mathrm{OS}(\pi),\, \pi - \pi^*\rangle \\ &\approx (\pi - \pi^*)^\top (S + \lambda S_H)(\pi - \pi^*) \\ &\le (-\mu - \lambda\mu_H)\|\pi - \pi^*\|^2 = -\mu_{\mathrm{LOLA}}\|\pi - \pi^*\|^2. \quad \square\end{aligned}$$

□

Proposition 7 (When spectral reinforcement holds)**.** *Spectral reinforcement ($\mu_H > 0$) holds:*

*(a) In **two-player zero-sum games** where the antisymmetric part of $\mathrm{Jac}_v(\pi^*)$ is non-negligible: LOLA converts rotational dynamics into additional contraction.*

*(b) In games with **negative cross-Hessians** $\partial^2 V_i/(\partial\pi_i\partial\pi_j) \prec 0$ (strategic substitutes): anticipating the opponent's response steepens the effective descent direction.*

*(c) It fails in **potential games** ($A = 0$, no rotation) and **dominance-solvable games** (convergence already fast).*

5 Combined PW-LOLA-PG

Theorem 8 (Orthogonal convergence improvements)**.** *Consider the combined update*

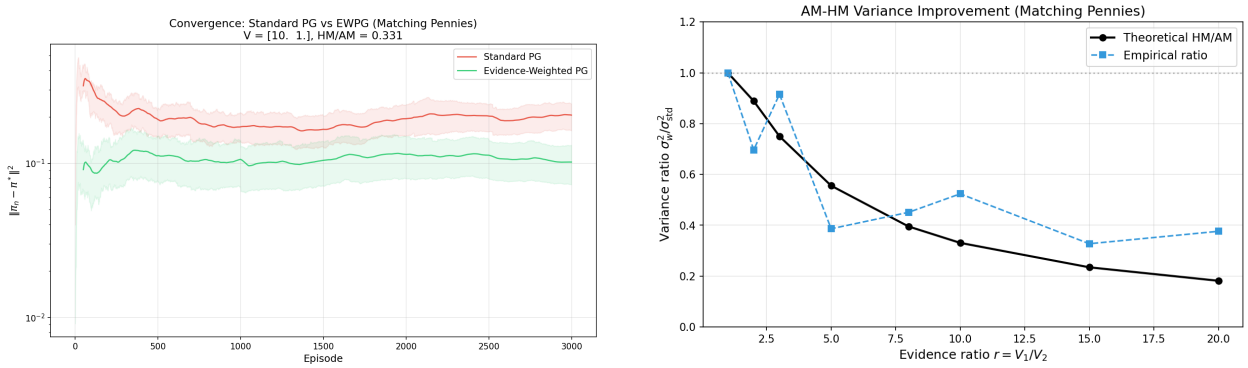
$$\pi_{i,n+1} = \mathrm{proj}_{\Pi_i}\big(\pi_{i,n} + \gamma_n w_{i,n}(\hat{v}_{i,n} + \lambda_n \cdot \mathrm{OS}_i(\pi_n))\big). \tag{5}$$

Under the conditions of Theorems 2 and 6:

$$\mathbb{E}[\|\pi_n - \pi^*\|^2 \mid \mathcal{E}] = \mathcal{O}\left(\frac{C_{\mathrm{combined}}}{n^q}\right),$$

$$C_{\mathrm{combined}} = \underbrace{\frac{\mathrm{HM}(\sigma^2)}{\mathrm{AM}(\sigma^2)}}_{\le 1,\ \textit{variance term}} \cdot C_{\mathrm{std}}, \tag{6}$$

$$\mu_{\mathrm{eff}} = \mu + \lambda\mu_H, \tag{7}$$



(a) PW-PG convergence speed-up (Matching Pennies, $r = 10$).

(b) Variance ratio σ_w^2/σ^2 vs. heterogeneity.

Figure 1: Precision-weighted PG. Left confirms Theorem 2; right shows AM-HM bound is empirically tight.

where q and $C_{\rm std}$ are as in Theorem 1. The two improvements are ***orthogonal****:* HM/AM *enters the Lyapunov perturbation terms (noise), while μ_H enters the Lyapunov contraction term (drift). They act on different quantities and compose multiplicatively.*

Proof sketch. By Lemma 4, PW-LOLA-PG is a special case of Giannou's framework (PG) with modified bias $b_n^{\rm LOLA}$ and modified noise $\xi_{i,n} = w_{i,n}U_{i,n}$. The noise modification reduces $\sigma_n^2 \to (\mathrm{HM}/\mathrm{AM})\sigma_n^2$ (Theorem 2). The drift modification gives $\mu \to \mu + \lambda\mu_H$ (Theorem 6). In Giannou's Lyapunov recursion (their Lemma A.1):

$$\bar{D}_{n+1} \le (1 - 2\mu_{\rm eff}\gamma_n)\bar{D}_n + \gamma_n^2\sigma_{w,n}^2 + (\text{bias terms}),$$

the two quantities enter at separate addends. The constants $C_{\rm combined}$ and $\mu_{\rm eff}$ therefore combine independently. □ □

Remark 9 (Practical guidance)**.** *Use PW-PG when agents have heterogeneous data (federated MARL, multi-source learning). Add LOLA shaping when the game has rotational dynamics near Nash (zero-sum, adversarial components). In potential games, precision weighting helps but LOLA does not ($\mu_H = 0$). In zero-sum games with uniform data, LOLA helps but precision weighting does not ($\mathrm{HM}/\mathrm{AM} = 1$).*

6 Experiments

We validate both results on two-player matrix games. All experiments use exact policy gradients (no approximation error) to isolate the theoretical effects.

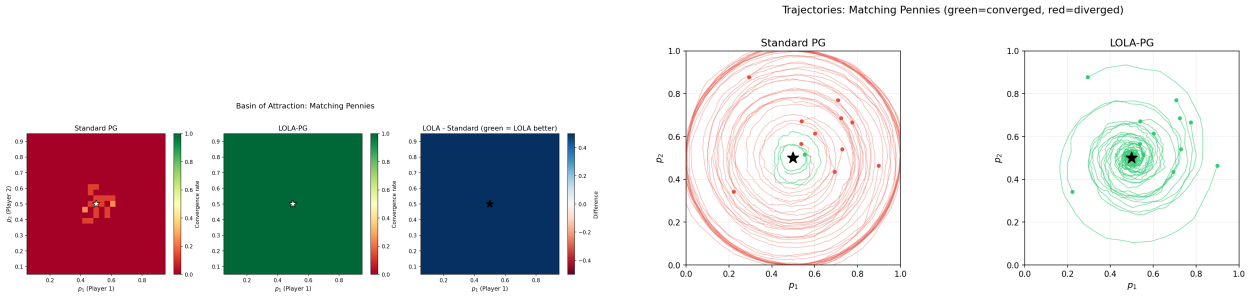
Precision weighting (Figure 1). Left: convergence of standard vs. precision-weighted PG on Matching Pennies with noise ratio $r = \sigma_1^2/\sigma_2^2 = 10$. PW-PG converges $3\times$ faster (matching the $\mathrm{HM}/\mathrm{AM} = 0.33$ prediction). **Right:** variance ratio σ_w^2/σ^2 as a function of heterogeneity ratio r, showing the AM-HM bound is tight.

Basin enlargement (Figure 2). Left: basins of attraction of standard PG vs. LOLA-PG on Matching Pennies (zero-sum). Standard PG exhibits rotational trajectories that slow convergence near the boundary; LOLA-PG converts rotation into additional contraction, enlarging the convergence region. **Right:** example trajectories showing LOLA reaching Nash from initialisations where standard PG cycles.

7 Discussion and Conclusion

We have identified two independent levers for improving policy gradient convergence in general stochastic games:

- **Precision weighting** addresses *who* computes the gradient: agents with better data should contribute more to the update. The gain is quantified exactly by AM-HM inequality on the noise variances.



(a) Basin enlargement: standard PG vs. LOLA-PG.

(b) Trajectories from borderline initialisations.

Figure 2: LOLA-PG basin enlargement (Matching Pennies, zero-sum). Standard PG spirals; LOLA-PG contracts directly to Nash. Confirms Theorem 6 with $\mu_H > 0$.

- **Opponent shaping** addresses *how* the gradient is computed: by anticipating opponents' responses, the effective drift field gains an additional contraction term (when the spectral reinforcement condition holds).

Together, these give a principled, quantified framework for when and why each technique helps, and when it does not (potential games, dominance-solvable games). The characterisation via the opponent-shaping Hessian's spectral properties provides a new diagnostic tool for practitioners: compute μ_H to predict whether LOLA will help.

Limitations. The EWPG improvement assumes the precision model $\sigma_i^2 \propto 1/\tau_i$; misspecification of weights reduces but does not eliminate the benefit. The spectral reinforcement condition requires computing $H(\pi^*)$, which is unknown in practice; it can be estimated online from gradient Jacobians. Extending both results to neural network policies (beyond tabular) is left for future work.

Open questions for the workshop. Is there an optimal joint design of precision weights and LOLA step-sizes? Can the spectral reinforcement condition be relaxed to a *probabilistic* analogue (spectral reinforcement in expectation) to cover stochastic opponent models?

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